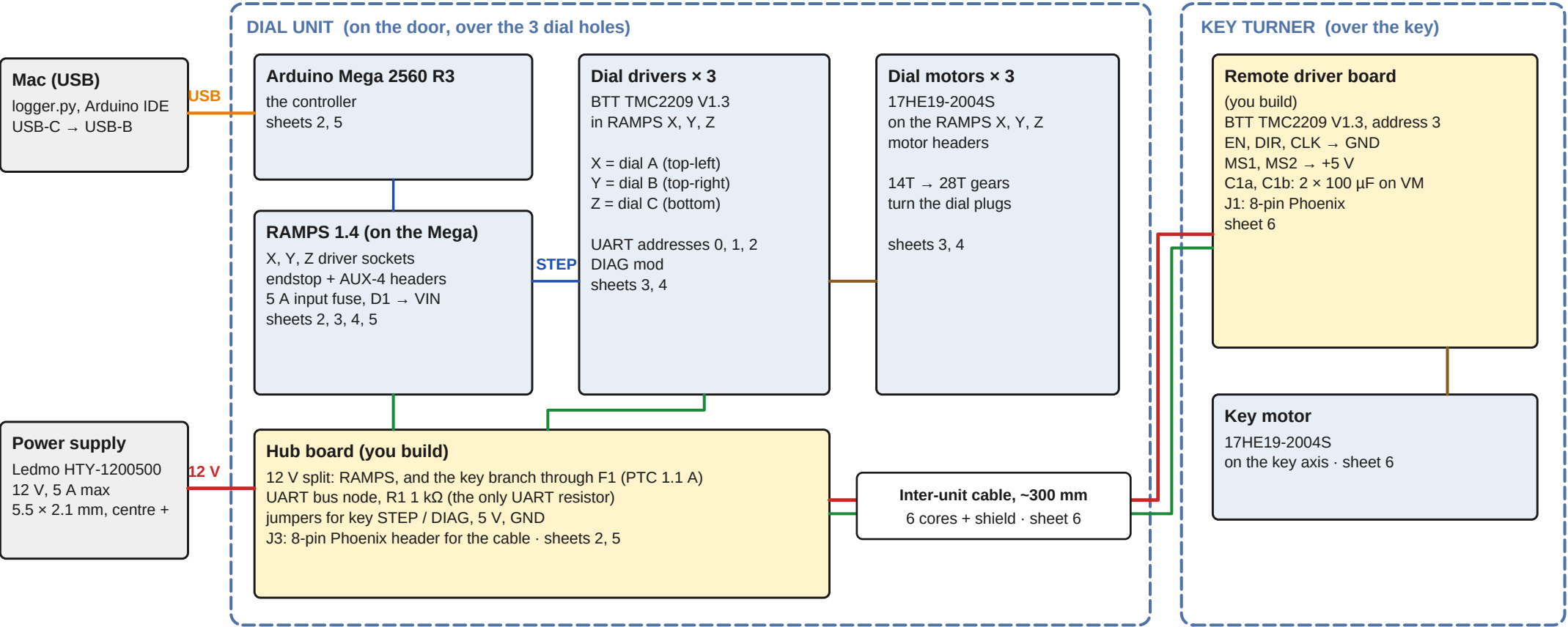


Overview

What connects to what. Each block is drawn in full on the sheet named in it.



Sheets

- 1 Overview (this sheet)
- 2 Power: 12 V, 5 V, GND
- 3 Dial drivers A and B on the RAMPS
- 4 Dial driver C; DIAG-mod and UART-tap details
- 5 Mega + RAMPS headers, hub board, UART bus
- 6 Inter-unit cable and key turner
- 7 Pin map, jumpers, currents, parts

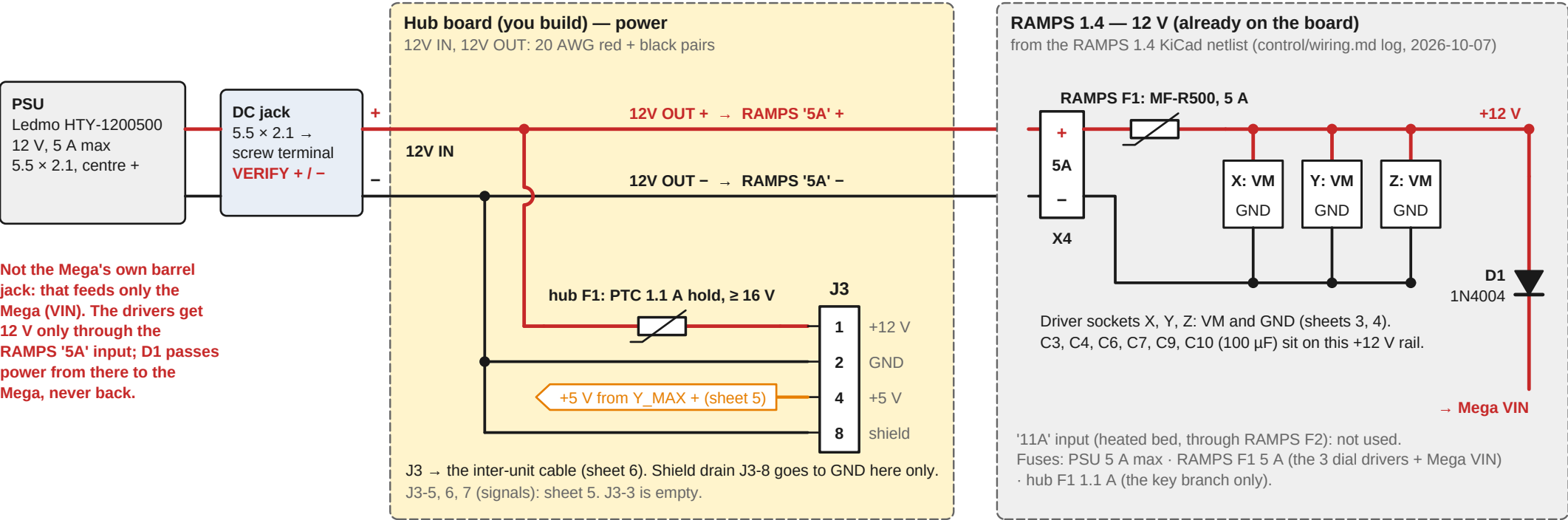
Rules

Make every connection with power off (USB unplugged, 12 V off).
Never plug or unplug a motor or the cable with 12 V on.
Power up: USB first, then 12 V. Power down: 12 V first.
Never fit an MS3 jumper: on the V1.3 that pin is the UART line.
Emergency stop: pull the 12 V plug. `!` aborts a move.
VERIFY = not yet checked on the bench (control/bringup.md).
Line colour = what a net does. Sheet 6 shows the cable's own colours.
If this disagrees with control/wiring.md, wiring.md wins.

- +12 V — +5 V (logic, VIO) — GND — UART bus
— STEP / DIR / EN, button — DIAG (stall) — motor coils — jumper / tie / spare
— bought module — board you build — already on the RAMPS / Mega: nothing to buy

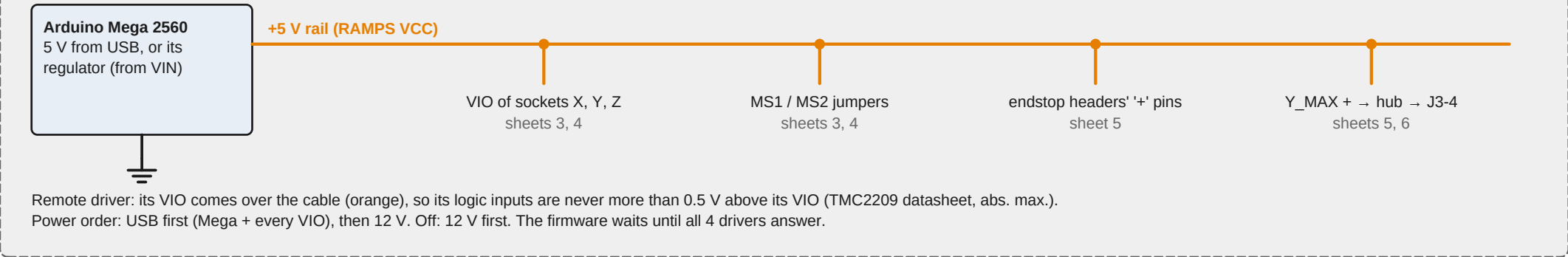
Power: 12 V, 5 V, GND

12 V comes in at the hub board and splits: the RAMPS (its own 5 A fuse) and the key branch (1.1 A PTC). 5 V logic comes from the Mega.



5 V logic (VIO) — from the Mega

VIO powers only each driver's I/O pins; its logic runs from its own VM (12 V)



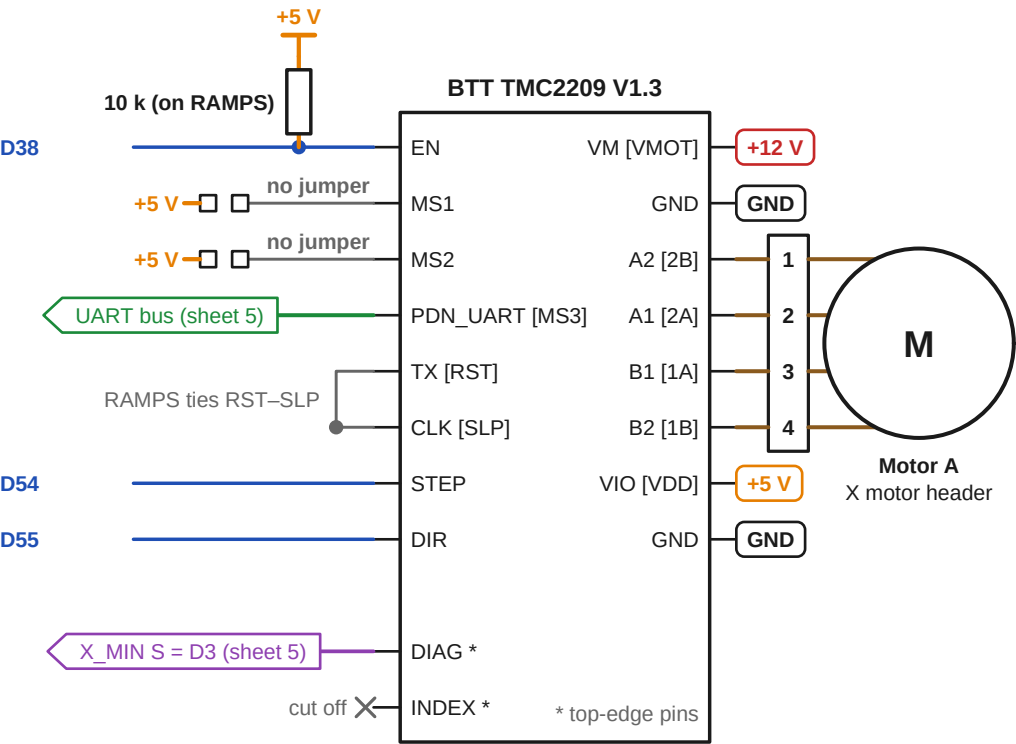
- +12 V
- +5 V (logic, VIO)
- GND
- UART bus
- STEP / DIR / EN, button
- DIAG (stall)
- motor coils
- jumper / tie / spare
- bought module
- board you build
- already on the RAMPS / Mega: nothing to buy

Dial drivers A and B on the RAMPS

[] = the RAMPS socket name of that pin. All connections to the left are on the RAMPS or by jumper.

Socket X — dial A (top-left)

UART address 0 · BTT TMC2209 V1.3 with the DIAG mod

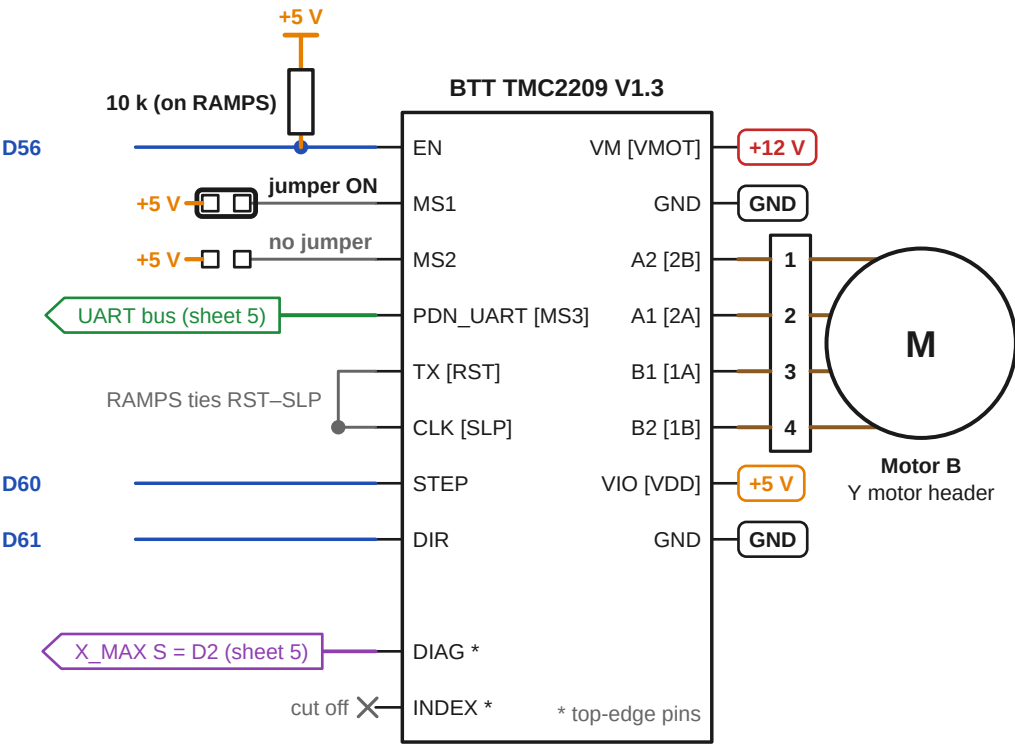


Jumpers under socket X: MS1 off, MS2 off, MS3 off (never fit MS3)

UART lead: clipped onto the MS3 jumper pin, signal side (sheet 4).
STEP D54 · DIR D55 · EN D38. The RAMPS pull-up keeps the driver off until the firmware pulls EN low.
DIAG lead (the mod, sheet 4) → X_MIN S pin: D3, an interrupt pin.
Motor 17HE19-2004S: header pins 1–2 = one coil, 3–4 = the other.

Socket Y — dial B (top-right)

UART address 1 · BTT TMC2209 V1.3 with the DIAG mod



Jumpers under socket Y: MS1 ON, MS2 off, MS3 off (never fit MS3)

UART lead: clipped onto the MS3 jumper pin, signal side (sheet 4).
STEP D60 · DIR D61 · EN D56. The RAMPS pull-up keeps the driver off until the firmware pulls EN low.
DIAG lead (the mod, sheet 4) → X_MAX S pin: D2, an interrupt pin.
Motor 17HE19-2004S: header pins 1–2 = one coil, 3–4 = the other.

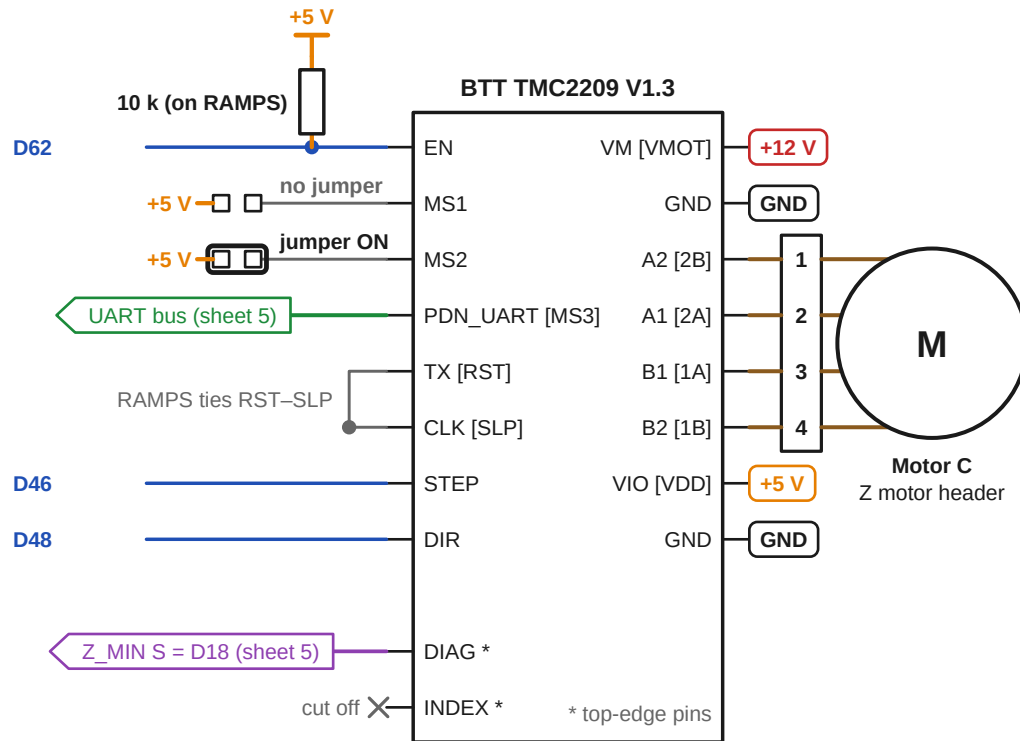
[] = the RAMPS / StepStick socket name of that pin. Fit each driver with EN, DIR, VM and GND matching the RAMPS silkscreen: a reversed driver is destroyed.
The BTT V1.3 puts PDN_UART in the socket's MS3 position. TX: R10 is not fitted, so TX connects to nothing. Don't bridge R10.
CLK: the RAMPS ties the socket's RST and SLP pins (= TX and CLK on the V1.3). With R10 open, CLK sees only its 20 k pull-down → internal clock.
Motor currents are set over UART: 1.0 A RMS, hold 0.5 A. If a motor buzzes instead of turning, swap the middle two wires in its plug.

- +12 V
- +5 V (logic, VIO)
- GND
- UART bus
- STEP / DIR / EN, button
- DIAG (stall)
- motor coils
- jumper / tie / spare
- bought module
- board you build
- already on the RAMPS / Mega: nothing to buy

Dial driver C; DIAG-mod and UART-tap details

Socket Z — dial C (bottom)

UART address 2 · BTT TMC2209 V1.3 with the DIAG mod



Jumpers under socket Z: MS1 off, MS2 ON, MS3 off (never fit MS3)

UART lead: clipped onto the MS3 jumper pin, signal side (sheet 4).

STEP D46 · DIR D48 · EN D62. The RAMPS pull-up keeps the driver off until the firmware pulls EN low.

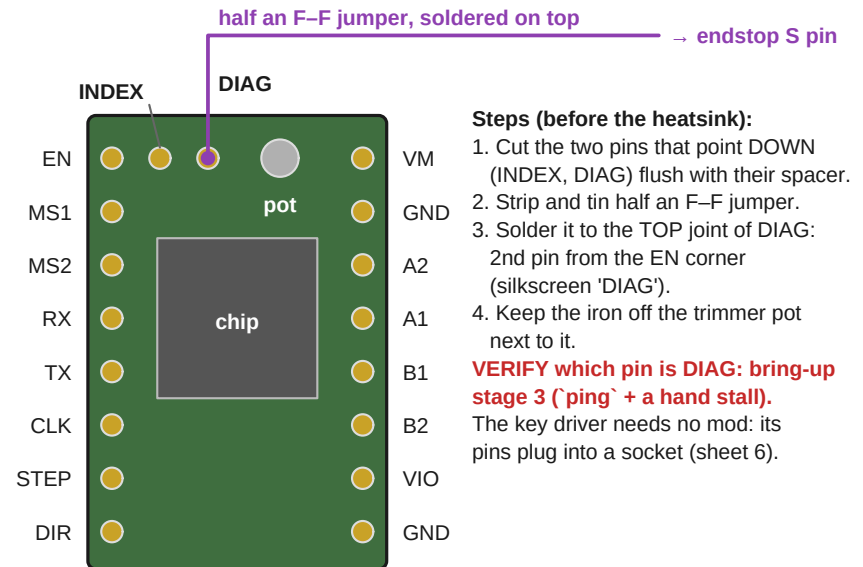
DIAG lead (the mod, sheet 4) → Z MIN S pin: D18, an interrupt pin.

Motor 17HE19-2004S: header pins 1-2 = one coil, 3-4 = the other.

— +12 V
 — +5 V (logic, VIO)
 — GND
 — UART bus
— STEP / DIR / EN, button
— DIAG (stall)
— motor coils
— jumper / tie / spare
 bought module
 board you build
 already on the RAMPS / Mega: nothing to buy

DIAG mod (the 3 dial drivers)

BTT TMC2209 V1.3 seen from above · not to scale



Steps (before the heatsink):

1. Cut the two pins that point DOWN (INDEX, DIAG) flush with their spacer.
2. Strip and tin half an F-F jumper.
3. Solder it to the TOP joint of DIAG:
2nd pin from the EN corner (silkscreen 'DIAG').
4. Keep the iron off the trimmer pot next to it.

VERIFY which pin is DIAG: bring-up stage 3 (‘ping’ + a hand stall).
The key driver needs no mod: its pins plug into a socket (sheet 6).

UART tap: the MS3 jumper pin (each socket)

RAMPS jumper block under a driver socket · not to scale



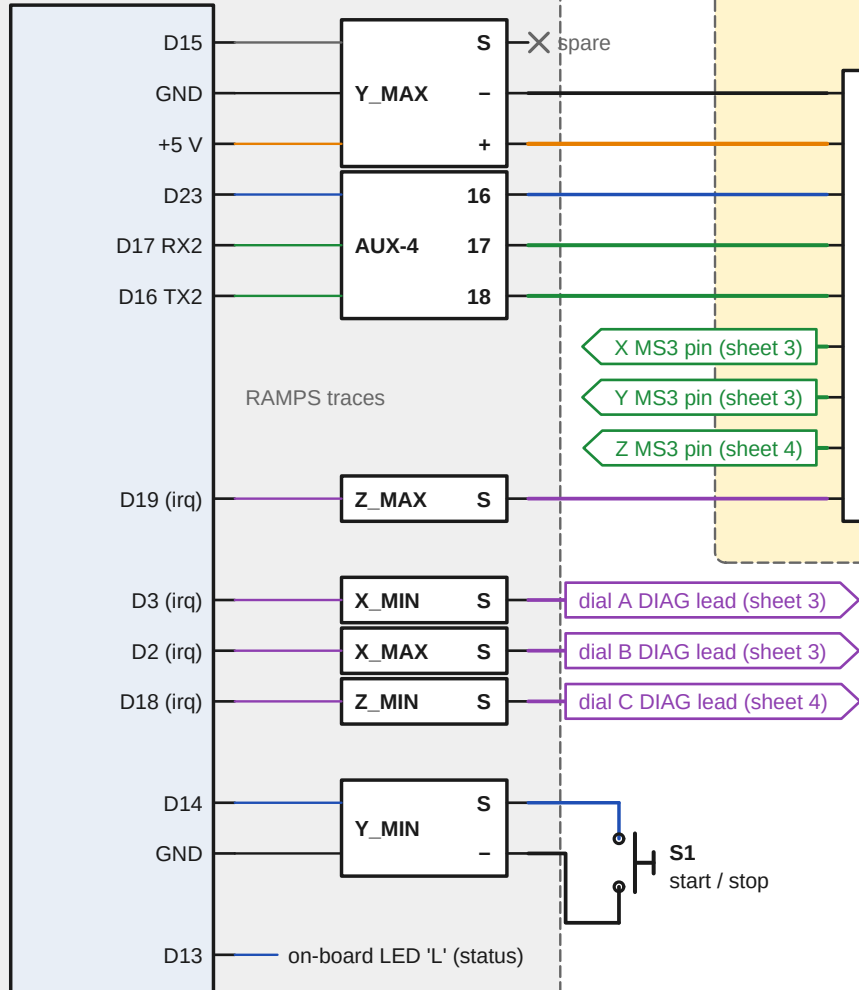
Mega + RAMPS headers, hub board, UART bus

Thin lines = RAMPS traces. Thick lines = your jumper wires (F–F, 10–20 cm) and hub-board wiring.

RAMPS 1.4 on the Mega

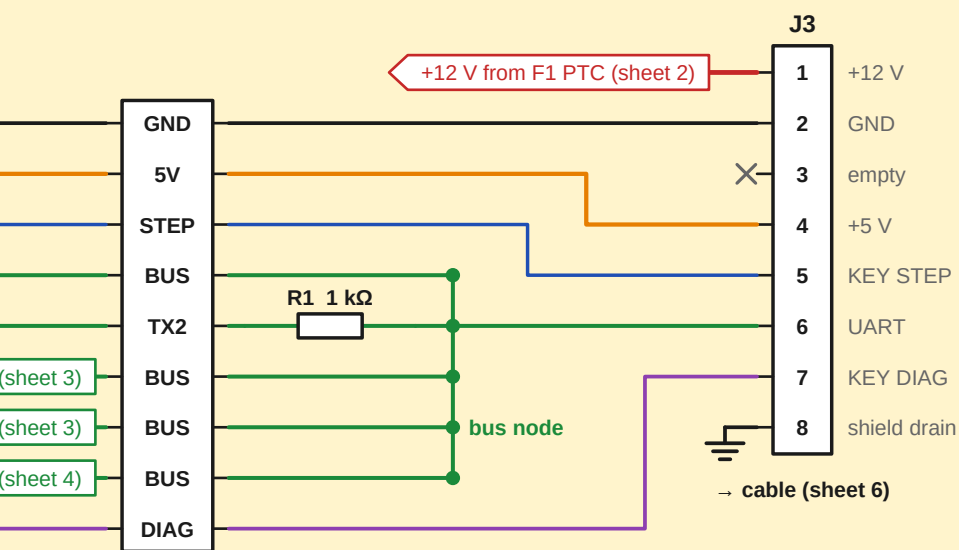
endstop headers: S signal, – GND, + 5 V

Mega 2560



Hub board (you build) — signals

2.54 mm male pins for the jumper ends; keep R1 right at the TX2 pin



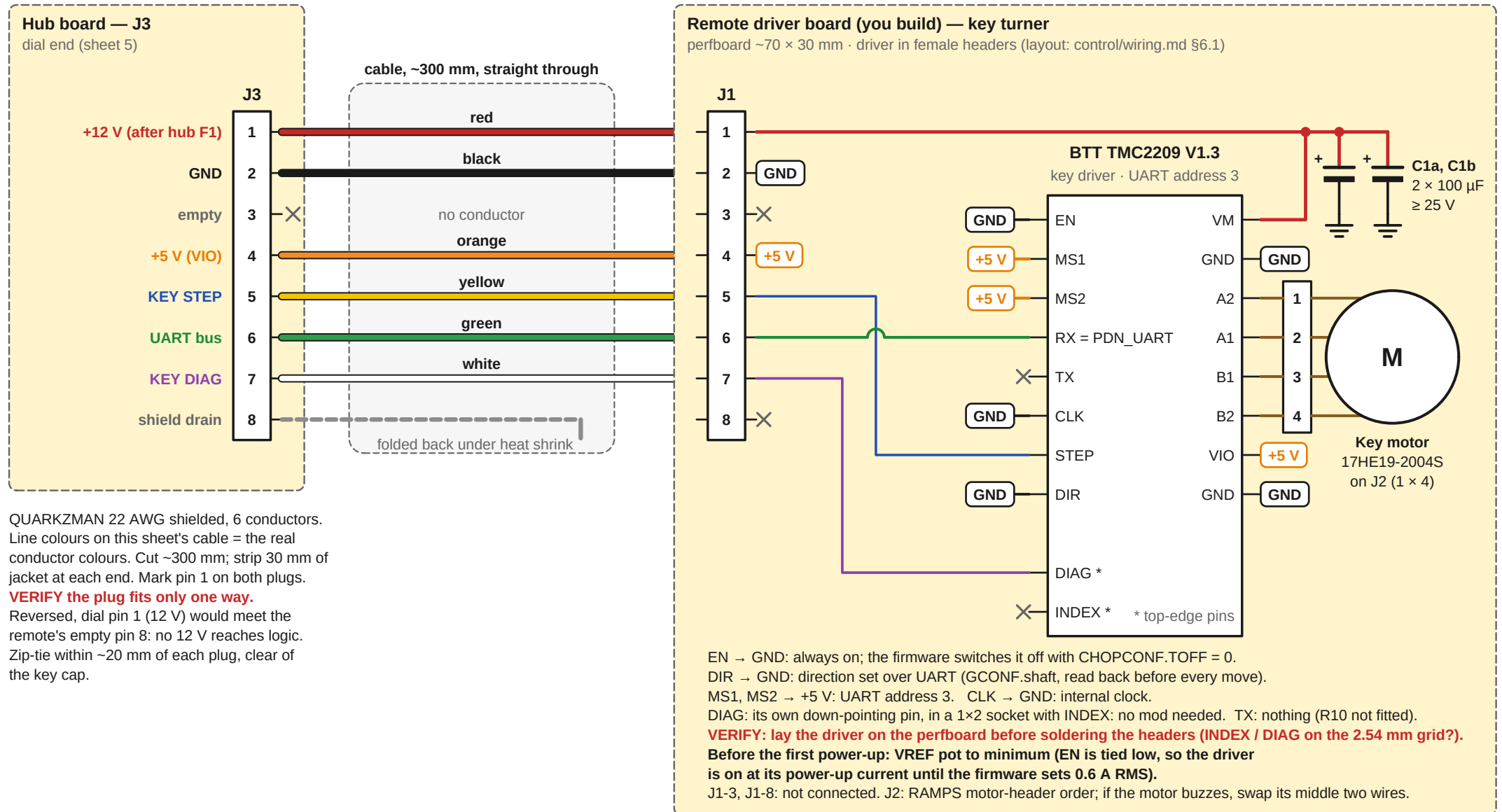
Notes

One 1 kΩ in total: TX2 → R1 → bus. RX2 and every driver's PDN_UART sit straight on the bus (TMC2209 datasheet Fig. 4.1). The Mega hears its own transmissions; TMCStepper skips that echo. SENDDELAY ≥ 2 on every driver. Addresses by MS1 / MS2: X 0, Y 1, Z 2, key 3. Bus at 115200 baud. (irq) = an interrupt pin. DIAG is a pulse: the firmware latches it with attachInterrupt on D2, D3, D18, D19 (Arduino interrupt numbers 0, 1, 5, 4). Inputs use the Mega's internal pull-ups: an unplugged DIAG reads high = 'stalled', so the firmware refuses to move (fail safe). Button S1 (Y_MIN): pauses a run; resumes or starts one when idle. Shield drain J3-8: GND at this end only. Hub GND = 12V IN – (sheet 2). The hub's 12 V parts (12V IN, 12V OUT, F1) are on sheet 2. STEP / DIR / EN of X, Y, Z: sheets 3, 4. AUX-4 15, 14 (D25, D27): reserved.



Inter-unit cable and key turner

Straight through: the same pin number at both ends. 12 V off before plugging or unplugging the cable.



Pin map, jumpers, currents, parts

Pin map (control/wiring.md §1; Marlin pins_RAMPS.h)

Function	Mega pin	Where it connects
Dial A STEP / DIR / EN	D54 / D55 / D38	RAMPS X socket
Dial B STEP / DIR / EN	D60 / D61 / D56	RAMPS Y socket
Dial C STEP / DIR / EN	D46 / D48 / D62	RAMPS Z socket
Dial A DIAG	D3 (interrupt)	X_MIN header, S
Dial B DIAG	D2 (interrupt)	X_MAX header, S
Dial C DIAG	D18 (interrupt)	Z_MIN header, S
Key DIAG (cable)	D19 (interrupt)	Z_MAX header, S, via the hub
Key STEP (cable)	D23	AUX-4 pin 16, via the hub
UART TX2	D16	AUX-4 pin 18 → hub R1
UART RX2	D17	AUX-4 pin 17 → hub bus
Dial drivers' PDN_UART	—	MS3 jumper pins X, Y, Z → hub bus
Start / stop button	D14 (pull-up)	Y_MIN header, S and –
5 V + GND for the remote	5 V, GND	Y_MAX header, + and –
Status LED	D13	on-board LED 'L'
Spare	D15	Y_MAX header, S
Reserved: key DIR, key EN	D25, D27	AUX-4 pins 15, 14
USB serial (the log)	D0 / D1	USB: keep free

Power order (§8)

1. Make every connection with power off (USB unplugged, 12 V off).
2. Before the first power-up: every driver's VREF pot to minimum.
3. Power up: USB first, then 12 V. 4. Power down: 12 V first, then USB.
5. Never plug or unplug a motor or the cable with 12 V on.
6. Emergency stop: pull the 12 V plug.

VERIFY on the bench (control/bringup.md)

Which MS3 jumper pin is the signal side (stage 1, `ping`).

Which top-edge pin is DIAG (stage 3). Phoenix plug keying.

Remote board: INDEX / DIAG on the 2.54 mm grid (before soldering headers).

Multimeter: DC-jack + / –; 5 V and 12 V polarity at the hub and remote board before plugging the driver in; cable continuity.

Sources: control/wiring.md — Marlin pins_RAMPS.h, TMC2209 datasheet rev 1.09, BTT TMC2209 V1.3 manual + schematic, RAMPS 1.4 KiCad netlist.

+12 V

+5 V (logic, VIO)

GND

UART bus

STEP / DIR / EN, button

DIAG (stall)

motor coils

jumper / tie / spare

bought module

board you build

already on the RAMPS / Mega: nothing to buy

Driver jumpers and UART addresses (§2)

Socket	Dial	MS1	MS2	MS3	Address
X	A (top-left)	off	off	off	0
Y	B (top-right)	ON	off	off	1
Z	C (bottom)	off	ON	off	2
remote	key	to 5 V	to 5 V	n/a	3

Never fit an MS3 jumper. Sockets E0 and E1 stay empty.

Motor currents, set over UART (§7)

Driver	Run	Hold	Why
Dials A, B, C	1.0 A RMS	0.5 A	until the dial torque is measured
Key	0.6 A RMS	0.3 A	then 2 × the measured minimum, ≤ 1.0 A
Firmware ceiling	1.2 A	—	BTT: active cooling above 1.2 A

Parts you fit (§5, §6, §10) · full list: docs/bom.md

Board	Ref	Part	Status
Hub	F1	PTC fuse ~1.1 A hold, ≥ 16 V (for now: 1.6 A glass fuse)	ordered
Hub	R1	1 kΩ: the only UART resistor (TX2 → bus)	have
Hub	J3	Phoenix-style 5.08 mm 8-pin header + plug	have
Hub	—	male pins × 9: GND 5V STEP TX2 BUS×4 DIAG	have
Hub	—	12V IN / 12V OUT: 20 AWG red + black	ordered
Remote	J1	Phoenix-style 5.08 mm 8-pin header + plug	have
Remote	C1a/b	2 × 100 μF, ≥ 25 V electrolytic, in parallel	ordered
Remote	J2	1 × 4 male header (key motor)	have
Remote	—	female headers 2 × (1 × 8) + 1 × (1 × 2)	have
Both	—	perfboard (kit): ~50 × 30 and ~70 × 30 mm	have
PSU	—	DC jack 5.5 × 2.1 mm → screw terminal	ordered
Wiring	—	F–F jumpers, ~20 × 10–20 cm	ordered
RAMPS	S1	7 mm push button (start / stop)	have

Fuses, diode, capacitors and resistors drawn in grey on sheets 2–4 (RAMPS F1, F2, D1, the six 100 μF, the 10 k pull-ups) are already on the RAMPS.